JINGXUAN MA

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Data Science M.S. student at Johns Hopkins with a background in Applied Statistics and data analytics. Skilled in R, advanced SQL, Python ETL pipelines using Apache Airflow, Tableau dashboard development, operations automation, stakeholder collaboration, and AI prompt engineering to accelerate data analysis.

EDUCATION

Johns Hopkins University | August 2025 - June 2027
Master of Science in Data Science | Data Science | Baltimore, US
University College Cork | September 2021 - June 2025
Bachelor of Science in Actuarial Sciences | Actuarial Sciences | Cork, Ireland | Beijing, China

PROFESSIONAL EXPERIENCE

AI Agent Intern | Beijing Yiling Network Technology Co., Ltd.   Beijing, China | June 2026 - Present
- Faced with producing JD-matched one-page resumes without breaking a locked Word template, built a Python/FastAPI + Next.js AI agent that ranks roles, extracts keywords, and injects tailored content into the master DOCX via OOXML edits
- Designed format-lock and quality-gate checks (shell fingerprint, hyperlink preservation, Word PDF one-page validation, evidence-linked bullets) so delivery copies keep master fonts, margins, and list styles without rebuilding the document
- Delivered JD-aware resume projections by scoring inventory entries on SQL/Tableau/Python keyword overlap, reordering skills, and preserving evidence links on every bullet

Data Analyst Intern | Yinhua Fund Management Co., Ltd.   Beijing, China | June 2023 - August 2023
- Given the need to support campaign review and market reporting, collected, organized, and cleaned market and customer data from multiple business materials, creating structured datasets for trend analysis and management communication
- Used Python, Excel, and visualization techniques to analyze market patterns, prepare charts and analytical materials, and convert raw performance data into insights that could support investment-product marketing decisions
- Prepared concise reporting assets for internal stakeholders by translating quantitative findings into business narratives, improving clarity around customer behavior, campaign performance, and market trends

Data Analyst Intern | Shenwan Hongyuan Securities Co., Ltd.   Beijing, China | June 2024 - August 2024
- Faced with the need to evaluate whether a pricing model library could meet both runtime and maintainability requirements, conducted a structured feasibility analysis across Python, pure & optimized C++, Eigen vectorization, OpenMP, and ctypes-based Python/C++ integration
- Built and benchmarked a 100,000-path Monte Carlo pricing simulation, applying compiler optimization, vectorized matrix operations, and multithreaded random-path generation to isolate major bottlenecks in computation-intensive pricing workflows
- Delivered a hybrid architecture recommendation that assigned heavy simulation and statistical computation to C++ while keeping configuration, preprocessing, and visualization in Python; reduced runtime from approx. 33.4s to approx. 1.4s with OpenMP optimization

PROJECTS

Insurance Claims Severity Modeling | Python, R, pandas, scikit-learn, Monte Carlo Simulation
- To estimate claim severity across heterogeneous policy segments, collected and cleaned historical claims data and engineered exposure- and policy-level features to support distributional analysis of loss costs
- Analyzed skewed loss distributions, outliers, and cost drivers; compared regression and tree-based approaches to evaluate trade-offs among predictive accuracy, robustness, and business interpretability for underwriting and reserving use cases
- Extended the modeling framework with Monte Carlo simulation and performance-benchmarking concepts to evaluate scalable pricing workflows, translating model outputs into risk segmentation, pricing review, and portfolio-level reporting insights

Tesla Vehicle Quality & Risk Analytics Pipeline | Python, Apache Airflow, SQL, Tableau, NHTSA API
- Built an end-to-end ETL pipeline extracting 11,349 NHTSA complaints and 60 recall actions across Tesla Models 3/S/X/Y (2015 2024); engineered dynamic model-name discovery and ID-based deduplication to reduce 30,777 raw records to 11,349 verified distinct complaints
- Engineered frequency- and severity-based risk indicators (crash/fire/injury-weighted scores by model-year and component) in SQL, applying an actuarial-style frequency ×severity loss framework to quantify and rank vehicle quality risk across product lines
- Built and published an interactive Tableau dashboard visualizing component-level risk rankings and recall-lag trends (first complaint to official recall), translating 11,349 complaint records into decision-ready risk insights for quality and safety stakeholders.

SKILLS & CERTIFICATIONS

Python, R, SQL, Tableau, Apache Airflow, data cleaning, feature engineering, exploratory analysis, Pandas, NumPy, scikit-learn, XGBoost, Monte Carlo Simulation, Credit Risk, Claims Modeling, stakeholder reporting